TP2/RR2 tests for American options on SPY (2025 dataset)

March 26, 2026

1 Summary results

The options data on SPY covers the period 1-Jan-25 to 29-Aug-25, for a total of 165 trading days.

Filters. Several conditions are applied in order to eliminate stale data.

1. **Volume filter.** Keep only options with non-zero trading volume.
2. **Non-zero bid filter.** Keep only options with non-zero bid price.
3. **Delta filter.** Keep only options with $|\Delta|$ in the range $[0.01, 0.99]$.

Table 1: The numbers of 4-uples in the data $n_{C,P}$, the violation counts $vio_{C,P}$ and the violation rates $vr_{C,P}$ for several data subsets.

Dataset	n_C	vio_C	vr_C	n_P	vio_P	vr_P
No Delta filter (mids)	67,115,004	3,823,583	5.697%	104,050,597	5,885,608	5.656%
Delta Filter (mids)	43,376,334	176,859	0.408%	62,305,648	1,182,236	1.897%
Delta Filter (bid-ask)	43,376,334	10,909	0.025%	62,305,648	232,228	0.373%

The violation counts and violation rates obtained for the entire dataset are shown in Table 1. The $\mathrm{TP}_2/\mathrm{RR}_2$ violations are computed using both mid-prices (second row) and using extreme bid-ask prices (third row).

For reference the first row shows the results obtained without applying the Delta filter, and mid prices. The Delta filter reduces the number of 4-uples and violations: its application leaves only 64.63% of the call 4-uples and only 59.88% of the put 4-uples. Furthermore, the Delta filter has an

asymmetric impact on calls and puts, as it eliminates more call TP_2 violations than put RR_2 violations.

Daily violation rates. Figure 1 shows the daily violation rates, separately for calls and puts. The dashed lines show the average violation rates.

During the period considered, the puts violation rates dominates over that of the calls, except for a short period. This is consistent with the average violation rate which is much larger for puts than for calls.

Distribution in maturity. The distribution of the violations in maturities (T_1, T_2) is shown as heat maps in Figure 2 aggregating over the entire dataset.

The grey dots show all traded (T_1, T_2) pairs. The colored dots show points with violation rates above a certain threshold $(1\%, 2\%, 5\%)$.

The results show that violations for calls are larger near the diagonal $T_1 = T_2$, regardless of maturity. This is somewhat different from the SPX European options, where violations were concentrated at short maturities.

The RR_2 violations show also concentration in a band close to the diagonal, and also in a vertical band with $T_1 \sim 50 - 100$ days, regardless of T_2. Figure 3 shows the distribution of the largest RR2 violations, up to 15%, which show also a concentration in a band of $T_1 \in [50, 100]$ days.

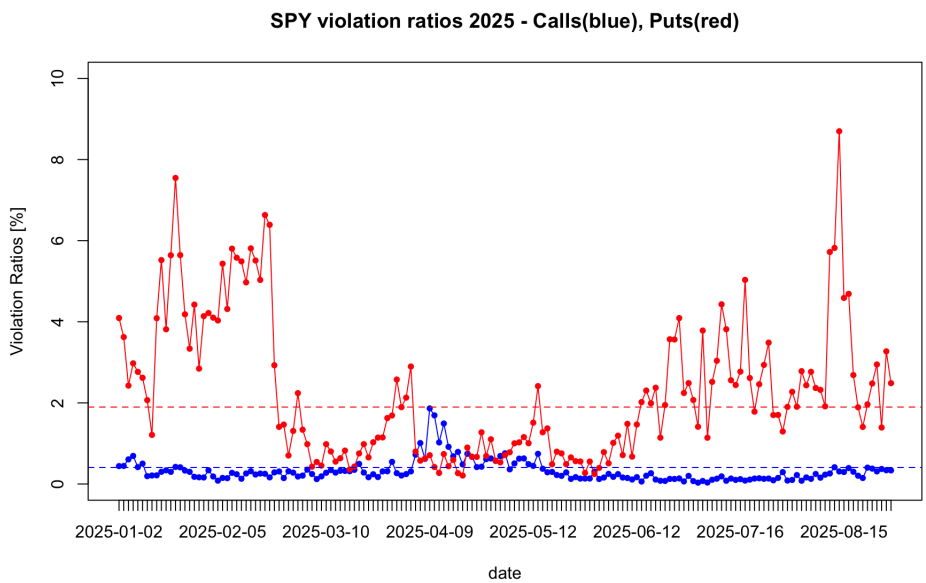

Figure 1: The violation rates for TP2/RR2 violations for each day in the 2025 dataset.

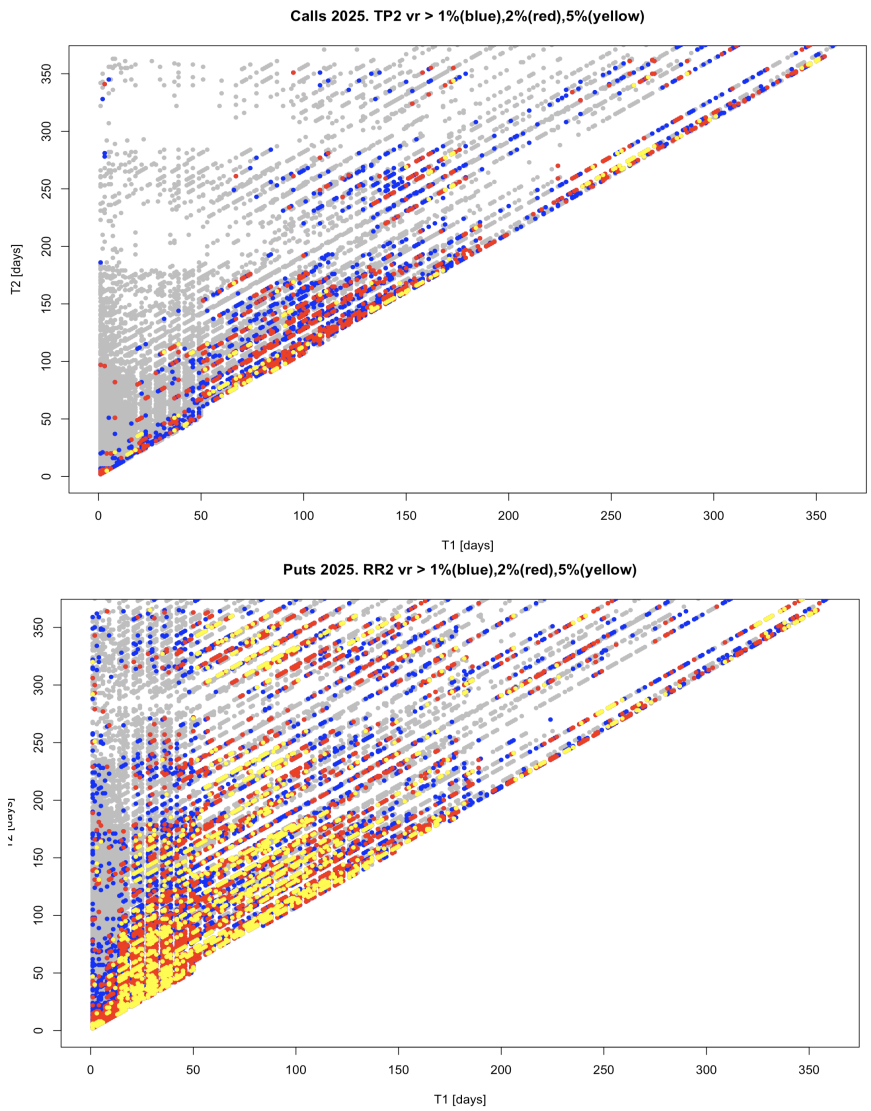

Figure 2: The distribution of TP2/RR2 violations in maturities (T_1, T_2), aggregating over all days in the 2025 dataset.

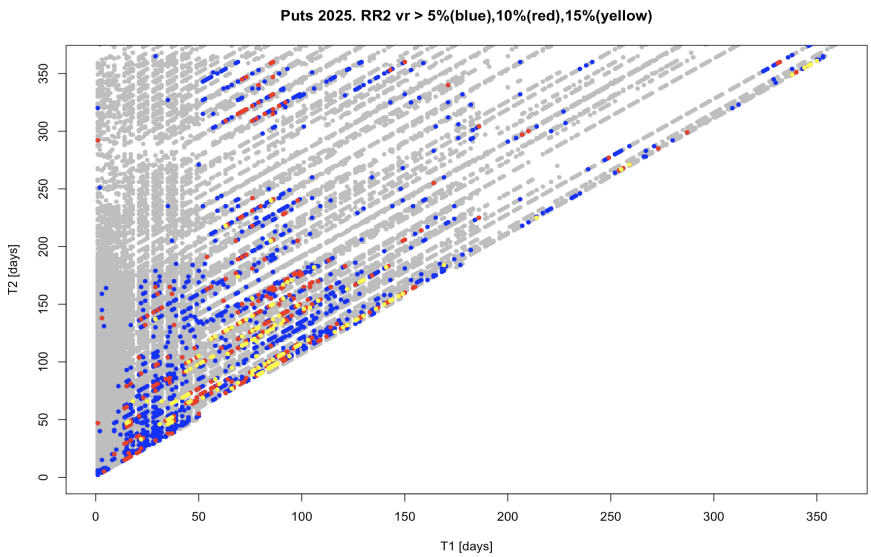

Figure 3: The distribution of largest RR2 violations in maturities (T_1, T_2), aggregating over all days in the 2025 dataset.